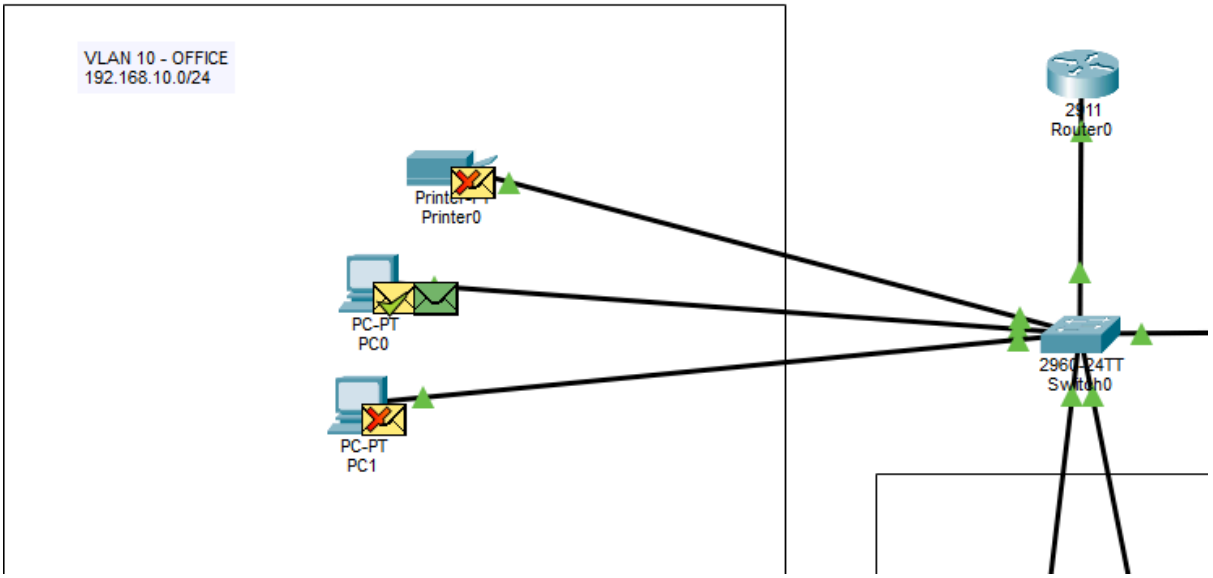
DHCP testing

Getting an IP address automatically initially failed,after trouble-shooting, I found the problem. The problem was that the extended ACLs configured on the sub-interfaces were causing the DHCP request to be dropped.

```
50 permit udp host 0.0.0.0 host 255.255.255.255 eq bootps (3 match(es))
60 permit udp host 0.0.0.0 host 255.255.255.255 eq bootpc
70 permit udp any any eq bootps
80 permit udp any any eq bootpc
```

These lines were added to each extended ACL to permit UDP from DHCP. After this, all devices successfully received the correct IP from the correct IP pools.



VLAN testing and ACL testing

Pinging VLAN 10s gateway from 192.168.10.2

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Pinging VLAN 20s gateway from 192.168.10.2

```
Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

The ACL runs as intended and blocks any traffic to another VLAN

SSH testing

192.168.10.2 should be able to SSH into 192.168.90.1

```
C:\> ssh -l admin 192.168.90.1

Password:

 No unauthorised access

R1_SSH>
```

```
C:\> ssh -l admin 192.168.90.1

Password:

 No unauthorised access

R1_SSH>exit

[Connection to 192.168.90.1 closed by foreign host]
C:\>ssh -l admin 192.168.20.1

% Connection timed out; remote host not responding
C:\>
```

192.168.10.2 should not be able to SSH into 192.168.20.1 due to the ACL that prevents VLANs from communicating with each other.

192.168.99.3 trying to SSH into 192.168.10.1 and 192.168.90.1

```
[Connection to 192.168.90.1 closed by foreign host]
C:\>ssh -l admin 192.168.10.1

% Connection timed out; remote host not responding
C:\>ssh -l admin 192.168.90.1

% Connection refused by remote host
C:\>
```

The ACL works as intended 192.168.99.3 should not be able to SSH into any interface.